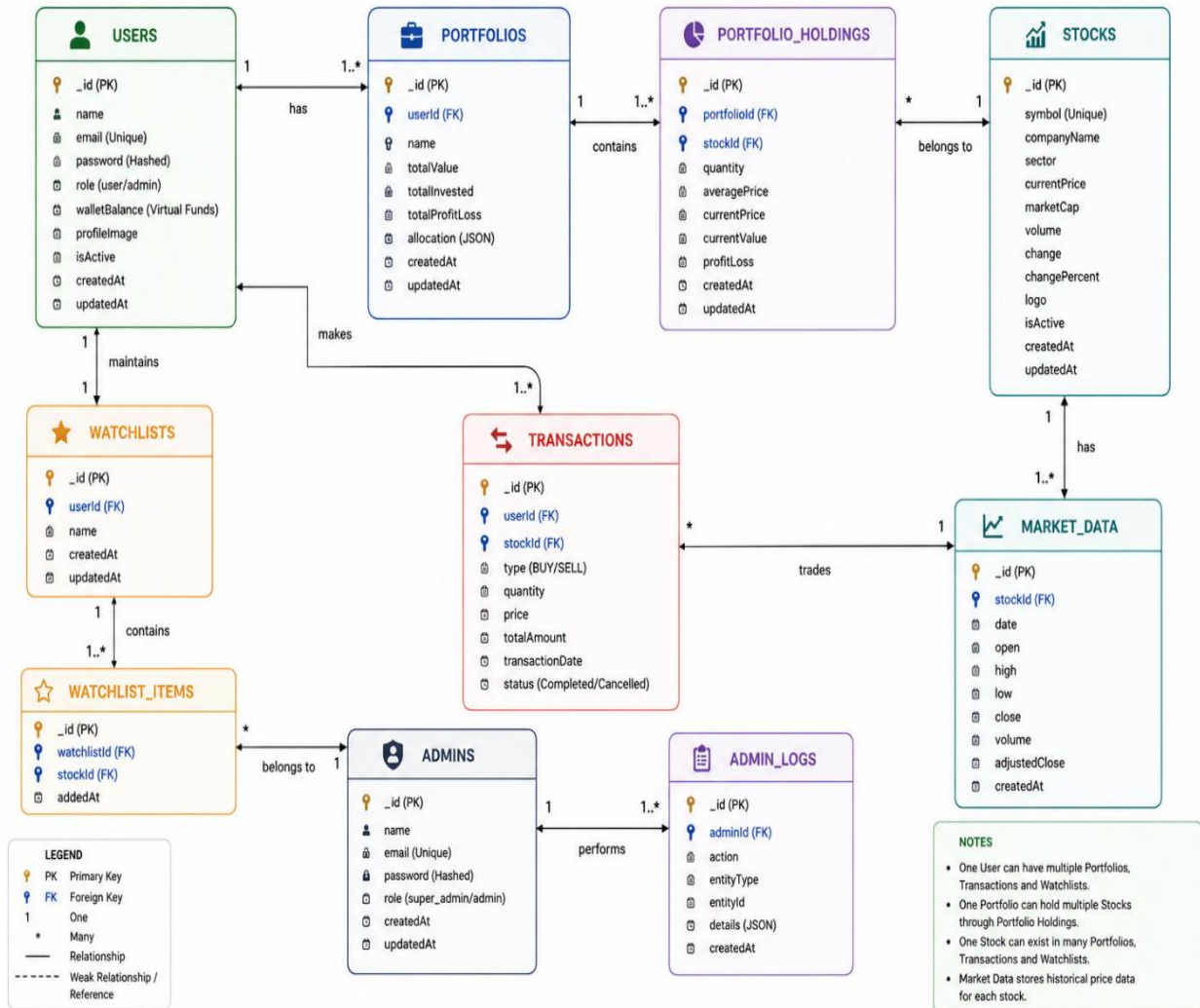


ER-DIAGRAM

SB STOCKS – ER DIAGRAM



1. USERS

This section acts as the primary anchor for the application's user profiles.

- **Core Attributes:** It tracks basic details like _id (Primary Key), name, email (enforced as Unique), and a secure password (noted as Hashed).
- **Financial & Identity Data:** It maintains a walletBalance field representing virtual funds available for paper trading, alongside a role (user/admin status) and profileImage.
- **System Controls:** Timestamps like createdAt and updatedAt are logged along with an isActive flag.

2. PORTFOLIOS & PORTFOLIO_HOLDINGS

This pair of tables structures user balances against asset allocations.

- **PORTFOLIOS:** A sub-module belonging to a user (userId as a Foreign Key). It keeps track of high-level financial health, such as totalValue, totalInvested, totalProfitLoss, and an allocation stored in a JSON object.
- **PORTFOLIO_HOLDINGS:** Because a single portfolio can hold multiple stocks, this collection acts as a breakdown ledger. It references the specific portfolioId and stockId, detailing the asset's active quantity, averagePrice at buy-in, dynamic currentPrice, currentValue, and immediate profitLoss.

3. STOCKS & MARKET_DATA

This segment defines the core financial assets tradable on the platform.

- **STOCKS:** Holds the data for each listing, identifying a stock by its _id, a unique symbol (e.g., AAPL), companyName, and financial identifiers like sector, currentPrice, marketCap, volume, and day-to-day changes (change and changePercent).
- **MARKET_DATA:** Handles historical time-series data for charting features. It points to a stockId and tracks OHLC metrics (open, high, low, close), trading volume, and adjustedClose over specific dates.

4. TRANSACTIONS

The central ledger records every trade executed on the platform.

- **Core Attributes:** It binds a user (userId) to a stock (stockId) at a specific moment.
- **Trade Specifics:** It captures the transaction type (enforcing BUY/SELL parameters), quantity, execution price, calculated totalAmount, exact transactionDate, and the current fulfillment status (Completed/Cancelled).

5. WATCHLISTS & WATCHLIST_ITEMS

This area tracks real-time user favorites without an ownership stake.

- **WATCHLISTS:** Creates a named tracker list (e.g., "Tech Stocks") linked to a user.
- **WATCHLIST_ITEMS:** Maps selected items, linking a watchlistId directly to a stockId to represent which company listings are pinned to that specific list.

6. ADMINS & ADMIN_LOGS

This independent cluster manages the administrative system and security tracking.

- **ADMINS:** Stores credentials and explicit access tiers for administrative entities, supporting roles like `super_admin` or `standard admin`.
- **ADMIN_LOGS:** To guarantee security audit compliance, this collection acts as an immutable ledger tracking administrative actions (`action`), identifying the affected entity type (`entityType`, `entityId`), and logging metadata inside a `details` (JSON) field.

Utility Blocks & Legends

- **LEGEND:** Translates visual cues across the model. It defines symbols for Primary Keys (yellow key), Foreign Keys (blue key), one-to-many indicators (1 to *), strong reference lines, and weak reference paths (dashed lines).
- **NOTES:** Explicitly highlights system constraints to guide developer implementation, outlining key relationships such as:
 - A single user maintaining multiple portfolios/watchlists.
 - Portfolios storing many stocks concurrently through holdings.
 - Market Data maintaining comprehensive, time-stamped pricing records.